

Solution Requirements and Technology Stack

FitBuddy AI Fitness Plan Generator

Concrete requirements, accepted inputs and the implementation stack.

Profile rules documented by the project

Field	Accepted input
Name	2 to 80 characters.
User ID	3 to 40 letters, numbers, underscores or hyphens.
Age and weight	Age 16 to 100; weight above 25 kg and up to 350 kg.
Goal	Weight loss, muscle gain, general wellness or flexibility.
Intensity	Low, medium or high; the interface labels these Easy, Steady and Focused.

Core acceptance checks

- A valid profile generates all seven days, a separate nutrition or recovery tip and a visible mode label.
- A repeated user ID updates the profile while plan generation creates a new saved plan record.
- Feedback changes the latest plan while the original remains available in plan history.
- A missing profile or plan returns a not-found response; invalid values produce validation feedback.

Technology stack

Layer	Project packages or tool
Web and API	FastAPI and Uvicorn.
Validation and settings	Pydantic and pydantic-settings; python-dotenv loads local settings.
Database	SQLAlchemy with SQLite.
Pages	Jinja2 templates, CSS and multipart form support.
AI integration	Google GenAI Python SDK (google-genai), optional for local demo use.

requirements.txt constrains major package versions. The README recommends Python 3.10 or newer, with Python 3.11 or 3.12 recommended.